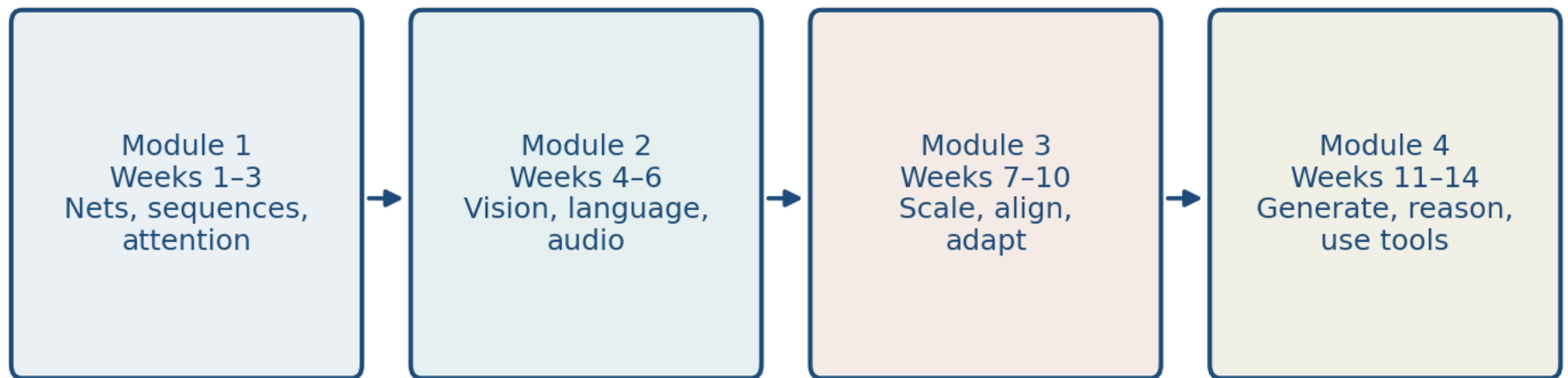


A self-contained LLM course

- DATA 441/641 and 442/642 are not prerequisites.
- Listed prereq: DATA 427/627.
- Overlap with those courses is intentional: more people can take this.
- The semester is a project. Lectures exist so that project has a backbone.

Where Week 1 sits: foundations before transformers



A language model is a stack

- Week 1 is the bottom: neurons, training, word vectors.
- Week 3 puts attention on those vectors.
- Later weeks align, retrieve, generate, and use tools.

A language model is a stack. Week 1 is the bottom two layers.

Applications: RAG, tools, multimodal products

Alignment and adaptation: SFT, RLHF, LoRA, retrieval

Transformer: self-attention over token embeddings

Token embeddings: distributed word (and later patch) vectors

This week: neurons, activations, gradient descent

The project starts this week

- A question you can measure, a public dataset, a baseline, one justified change.
- Undergraduate: group report and talk. Graduate: main project plus a paper mini-project.
- Lab 1 ends with a five-line topic seed. You may change it.